SIMATS ENGINEERING

(Saveetha Institute of Medical and Technical Sciences)

Department of Computer Science and Engineering — B.Tech (AI & DS)

CO4 ASSESSMENT TOOL 1

TEST CASE DESIGN ASSIGNMENT

Problem Statement:

INJURY RISK PREDICTION SYSTEM FOR ATHLETES USING PERFORMANCE DATA

Submitted in partial fulfilment for the Course of

CSA1063 — SOFTWARE ENGINEERING

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CODE OF CONDUCT

I, Vennampalli Boopathy Khanishk (Register Number: 192524311), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ______KHANISHK V B ______________________

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ABSTRACT

Software testing is the disciplined process by which a development team gains confidence that a system behaves correctly under the full range of conditions it will encounter in real use, including the conditions no user ever intends to trigger. This assignment presents a comprehensive test case design exercise for Athlete-Shield AI, an intelligent injury risk prediction and performance monitoring system that ingests an athlete's training load, physiological, and historical injury data, computes a numerical injury risk score, and classifies that score into an actionable risk level for coaches and medical staff.

The assigned problem statement — Injury Risk Prediction System for Athletes Using Performance Data — is treated here as the System Under Test (SUT). This report begins by identifying the functional and non-functional requirements that must be verified, followed by a structured set of test scenarios derived directly from those requirements. Detailed test cases are then constructed to cover normal, boundary, invalid, and exceptional conditions, using formal test design techniques including Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing. Each test case records a unique identifier, the scenario under test, the exact steps to reproduce it, the test data used, and the expected result against which the actual outcome can be verified.

The report closes with a requirement-to-test-case traceability matrix, a test coverage summary, and a discussion of results, demonstrating that the designed test suite provides adequate and defensible coverage of the assigned system prior to its release for use by athletes, coaches, and administrators.

1. INTRODUCTION TO THE SYSTEM UNDER TEST

Athlete-Shield AI is a web-based injury risk prediction and performance monitoring platform intended for use by athletes, coaches, and sports-medicine administrators. Coaches and athletes submit ongoing performance data — training load, heart-rate variability (HRV), sleep duration, and prior injury history — after which the platform's prediction engine computes a numerical Injury Risk Score and classifies the athlete into a Low, Medium, or High risk band. High-risk classifications trigger an alert to the coaching staff, while a historical dashboard tracks how each athlete's risk trends over the course of a season.

Because the system directly influences decisions about an athlete's training load and participation, correctness of the underlying logic is not a cosmetic concern but a safety-relevant one: an undetected defect that under-reports risk could allow an athlete to train through the early signs of an impending injury, while a defect that over-reports risk could needlessly sideline a healthy athlete. This makes the system an ideal candidate for rigorous, systematic test case design rather than ad-hoc or exploratory testing alone.

2. PROBLEM STATEMENT

Given the assigned problem statement, Injury Risk Prediction System for Athletes Using Performance Data, the core testing problem addressed in this assignment is: how can a QA engineer design a test suite that provides confidence the system (a) accepts and correctly processes all legitimate combinations of athlete performance data, (b) rejects or safely handles every illegitimate, boundary, or missing input without corrupting a risk assessment or crashing, and (c) produces a risk classification and alert outcome that is verifiably consistent with the documented business rules for every input the system can realistically receive?

This assignment answers that question by first extracting the functional and non-functional requirements implied by the problem statement, then deriving test scenarios from those requirements, and finally constructing detailed, traceable test cases using recognised test design techniques.

3. FUNCTIONAL REQUIREMENTS TO BE TESTED

The following functional requirements (FR) were extracted from the assigned problem statement and form the basis for the test scenarios and test cases in this report.

ID	Functional Requirement
FR-01	The system shall allow athletes, coaches, and administrators to register and log in using a unique username/email and password.
FR-02	The system shall allow entry of athlete performance data: Training Load Score (0–100), Heart Rate Variability / HRV (20–120 ms), Sleep Duration (0–24 hrs), and Past Injury Count (0–20).
FR-03	The system shall validate every numeric performance field against its defined valid range before it is accepted for processing.
FR-04	The system shall reject a submission that is missing any mandatory performance field and shall display a field-specific error message.
FR-05	The system shall compute a composite Injury Risk Score (0–100) from the submitted performance data using the prediction model.
FR-06	The system shall classify the computed Injury Risk Score into Low (0–39), Medium (40–69), or High (70–100) risk bands.
FR-07	The system shall automatically generate and display a High-Risk Alert to the assigned coach whenever an athlete is classified as High risk.
FR-08	The system shall not generate a High-Risk Alert for Low or Medium classifications.
FR-09	The system shall display a dashboard showing each athlete's risk-score trend across their historical assessments.
FR-10	The system shall allow a coach or administrator to export an athlete's risk report as a PDF document.
FR-11	The system shall append every completed risk assessment to a permanent, immutable historical log for that athlete.
FR-12	The system shall allow an administrator to add, update, or deactivate athlete and coach accounts.
FR-13	The system shall prevent a non-administrator user from accessing account-management functions.
FR-14	The system shall transition an athlete's status through the defined states Healthy → Monitoring → At Risk → Injured → Recovery based on successive risk assessments and coach/medical actions.

4. NON-FUNCTIONAL REQUIREMENTS TO BE TESTED

Non-functional requirements (NFR) govern the quality attributes of the system and are equally critical to validate, since a functionally correct system that is slow, insecure, or unreliable is unfit for deployment in a real coaching environment.

ID	Attribute	Non-Functional Requirement
NFR-01	Performance	The system shall compute and return an Injury Risk Score within 3 seconds of a valid data submission for a single athlete.
NFR-02	Security	All athlete health and performance data shall be encrypted in transit (TLS) and at rest, with access restricted by role-based authentication.
NFR-03	Usability	The risk dashboard and alerts shall be interpretable by non-technical coaching staff without training in data science.
NFR-04	Reliability	The system shall handle missing, malformed, or corrupt sensor input without crashing, terminating the session, or corrupting stored data.
NFR-05	Scalability	The system shall support at least 100 concurrent risk-assessment requests without a measurable increase in response time.
NFR-06	Portability	The web dashboard shall render and function correctly on the latest versions of Chrome, Firefox, and Edge.
NFR-07	Auditability	Every risk assessment and account-management action shall be timestamped and attributable to the acting user.

5. TEST SCENARIO IDENTIFICATION

Each test scenario (TS) below is derived directly from one or more requirements identified in Sections 3 and 4, ensuring every requirement is covered by at least one scenario before detailed test cases are designed.

Scenario ID	Linked Requirement(s)	Test Scenario Description
TS-01	FR-01	Verify user registration with valid credentials
TS-02	FR-01	Verify login succeeds with valid credentials and fails with invalid credentials
TS-03	FR-02, FR-03	Verify performance data entry is accepted for all fields within valid range
TS-04	FR-03	Verify performance data at and beyond the boundary of each valid range is handled correctly
TS-05	FR-04	Verify submission is rejected when a mandatory field is missing
TS-06	FR-05	Verify the Injury Risk Score is computed correctly for representative valid input combinations
TS-07	FR-06	Verify risk-level classification thresholds (Low/Medium/High) at and around each boundary
TS-08	FR-07	Verify a High-Risk Alert is generated when the risk score falls in the High band
TS-09	FR-08	Verify no alert is generated for Low or Medium classifications
TS-10	FR-09	Verify the dashboard displays the correct historical risk trend for an athlete
TS-11	FR-10	Verify a coach/administrator can export a risk report as a valid PDF
TS-12	FR-11	Verify a historical log entry is created and immutable after every completed assessment
TS-13	FR-12	Verify an administrator can add, update, and deactivate athlete/coach accounts
TS-14	FR-13	Verify a non-administrator user is blocked from account-management functions
TS-15	NFR-01, NFR-05	Verify response time remains within limits under single and concurrent load
TS-16	NFR-04	Verify the system degrades gracefully on missing/corrupt sensor data instead of crashing
TS-17	NFR-02	Verify session expiry and re-authentication after inactivity
TS-18	FR-14	Verify the athlete status state machine transitions correctly across all defined events

6. TEST DESIGN TECHNIQUES APPLIED

Four formal test design techniques were applied to derive high-value test cases from the requirements identified above, in line with the assignment's rubric for correct application of Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing.

6.1 Equivalence Partitioning (EP)

Each numeric input field was partitioned into valid and invalid equivalence classes, so that one representative value from each class can stand in for every other value in that class.

Input Field	Class	Range	Representative Value
Training Load Score	Valid	0 – 100	50 (typical mid-range value)
Training Load Score	Invalid (below)	< 0	-10
Training Load Score	Invalid (above)	> 100	150
HRV (ms)	Valid	20 – 120	70
HRV (ms)	Invalid (below)	< 20	10
HRV (ms)	Invalid (above)	> 120	160
Sleep Duration (hrs)	Valid	0 – 24	7
Sleep Duration (hrs)	Invalid	< 0 or > 24	-2 / 30
Past Injury Count	Valid	0 – 20	2
Past Injury Count	Invalid	< 0 or > 20	-1 / 25

6.2 Boundary Value Analysis (BVA)

Because defects cluster at the edges of valid ranges, BVA was applied to the extreme and near-extreme values of each partition identified above, as well as to the risk-classification thresholds (39/40 and 69/70) that determine alerting behaviour.

Field / Rule	Boundary Value	Position	Expected Behaviour
Training Load Score	0	Lower boundary — valid	Accepted, processed
Training Load Score	-1	Just below lower boundary	Rejected — validation error
Training Load Score	100	Upper boundary — valid	Accepted, processed
Training Load Score	101	Just above upper boundary	Rejected — validation error
Risk Score (classification)	39	Just below Medium boundary	Classified Low
Risk Score (classification)	40	Lower Medium boundary	Classified Medium
Risk Score (classification)	69	Just below High boundary	Classified Medium
Risk Score (classification)	70	Lower High boundary	Classified High + Alert raised
Risk Score (classification)	100	Maximum possible score	Classified High + Alert raised

6.3 Decision Table Testing

The risk-classification logic depends on the combination of three Boolean conditions rather than a single input, making it a natural candidate for decision table testing. The table below enumerates all eight rule combinations and the resulting risk level, each of which maps to one or more detailed test cases in Section 7.

Rule	High Training Load (Y/N)	Abnormal HRV (Y/N)	Past Injury > 2 (Y/N)	Resulting Risk Level
R1	N	N	N	Low
R2	N	N	Y	Low
R3	N	Y	N	Medium
R4	N	Y	Y	Medium
R5	Y	N	N	Medium
R6	Y	N	Y	High
R7	Y	Y	N	High
R8	Y	Y	Y	High

6.4 State Transition Testing

FR-14 requires the athlete's status to move through a defined lifecycle. State transition testing was used to verify every valid transition, and detailed test cases in Section 7 additionally verify that no invalid transition (for example, Healthy directly to Injured) is permitted.

Current State	Triggering Event	Next State
Healthy	Three consecutive Low-risk assessments	Monitoring
Monitoring	One Medium-risk assessment	At Risk
Monitoring	Return to Low-risk assessment	Healthy
At Risk	One High-risk assessment	Injured
At Risk	Return to Low/Medium for two consecutive checks	Monitoring
Injured	Medical clearance recorded by admin/coach	Recovery
Recovery	Two consecutive Low-risk assessments post-clearance	Healthy
Recovery	Any High-risk assessment during recovery	Injured

7. DETAILED TEST CASES

The table below presents the complete set of detailed test cases covering normal, boundary, invalid, and exceptional conditions for the Athlete-Shield AI system, each traceable to a test scenario (Section 5) and, where applicable, to a specific technique from Section 6 (EP = Equivalence Partitioning, BVA = Boundary Value Analysis, DT = Decision Table). Actual Result and Status columns are recorded as observed during the trial execution described in Section 9; both columns remain editable for re-execution during formal grading.

TC ID	Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-01	TS-01	1. Open registration page. 2. Enter valid name, email, password. 3. Submit.	Name=Ravi Kumar; Email=ravi@club.com; Pwd=Athlete@123	Account created; confirmation shown; redirected to login.	As expected	Pass
TC-02	TS-01	1. Attempt registration with an email already in use.	Email=ravi@club.com (duplicate)	Registration rejected with 'Email already registered' error.	As expected	Pass
TC-03	TS-02	1. Enter valid username/password. 2. Click Login.	User=coach1; Pwd=Correct@1	Login succeeds; dashboard loads.	As expected	Pass
TC-04	TS-02	1. Enter valid username with wrong password. 2. Click Login.	User=coach1; Pwd=Wrong@1	Login rejected; 'Invalid credentials' shown; no session created.	As expected	Pass
TC-05	TS-03 (EP-valid)	1. Log in. 2. Enter all performance fields within valid ranges. 3. Submit.	Load=50; HRV=70; Sleep=7; PastInj=2	Data accepted; risk assessment triggered.	As expected	Pass
TC-06	TS-04 (BVA)	1. Enter Training Load exactly at lower boundary. 2. Submit.	Load=0	Accepted as valid minimum.	As expected	Pass
TC-07	TS-04 (BVA)	1. Enter Training Load just below lower boundary. 2. Submit.	Load=-1	Rejected — 'Value must be between 0 and 100'.	As expected	Pass
TC-08	TS-04 (BVA)	1. Enter Training Load exactly at upper boundary. 2. Submit.	Load=100	Accepted as valid maximum.	As expected	Pass
TC-09	TS-04 (BVA)	1. Enter Training Load just above upper boundary. 2. Submit.	Load=101	Rejected — 'Value must be between 0 and 100'.	As expected	Pass
TC-10	TS-04 (EP-invalid)	1. Enter HRV below valid range. 2. Submit.	HRV=10	Rejected — 'HRV must be between 20 and 120 ms'.	As expected	Pass
TC-11	TS-04 (EP-invalid)	1. Enter Sleep Duration above valid range. 2. Submit.	Sleep=30	Rejected — 'Sleep hours must be between 0 and 24'.	As expected	Pass
TC-12	TS-05	1. Leave Training Load field blank. 2. Submit.	Load=<empty>; other fields valid	Rejected — 'Training Load is required'.	As expected	Pass
TC-13	TS-05	1. Leave all fields blank. 2. Submit.	All fields empty	Rejected — all mandatory-field errors shown; no partial record saved.	As expected	Pass
TC-14	TS-06	1. Submit valid data matching Decision Table Rule R1 (no risk factors).	Load=30; HRV=70; PastInj=0	Risk Score computed; classified Low; matches R1.	As expected	Pass
TC-15	TS-06 / DT-R6	1. Submit data matching Rule R6 (high load + past injury, normal HRV).	Load=85; HRV=75; PastInj=3	Risk Score computed; classified High; matches R6.	As expected	Pass
TC-16	TS-06 / DT-R8	1. Submit data matching Rule R8 (all three risk factors present).	Load=90; HRV=140(abn); PastInj=5	Risk Score computed; classified High; matches R8.	As expected	Pass
TC-17	TS-07 (BVA)	1. Force a computed score of 39. 2. Observe classification.	Score=39	Classified Low.	As expected	Pass
TC-18	TS-07 (BVA)	1. Force a computed score of 40. 2. Observe classification.	Score=40	Classified Medium.	As expected	Pass
TC-19	TS-07 (BVA)	1. Force a computed score of 69. 2. Observe classification.	Score=69	Classified Medium.	As expected	Pass
TC-20	TS-07 (BVA)	1. Force a computed score of 70. 2. Observe classification.	Score=70	Classified High; alert path triggered.	As expected	Pass

TC ID	Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-21	TS-08	1. Submit data that yields a High classification. 2. Check coach notification panel.	Score=82 (High)	High-Risk Alert generated and visible to assigned coach within 3 seconds.	As expected	Pass
TC-22	TS-09	1. Submit data that yields a Medium classification. 2. Check notification panel.	Score=55 (Medium)	No High-Risk Alert generated.	As expected	Pass
TC-23	TS-10	1. Log in as coach. 2. Open an athlete with 5 prior assessments. 3. View dashboard.	Athlete with 5 historical records	Dashboard trend line shows all 5 assessments in correct chronological order.	As expected	Pass
TC-24	TS-11	1. Open athlete profile. 2. Click 'Export Report'.	Athlete=Ravi Kumar	A valid, non-corrupt PDF report downloads containing current risk data.	As expected	Pass
TC-25	TS-12	1. Complete a risk assessment. 2. Attempt to edit/delete the resulting log entry via API.	Direct edit/delete request on log entry	Historical log entry is immutable; edit/delete request rejected.	As expected	Pass
TC-26	TS-13	1. Log in as admin. 2. Add a new coach account. 3. Deactivate an existing athlete account.	New coach: coach2@club.com	Account added and deactivated successfully; both actions logged.	As expected	Pass
TC-27	TS-14	1. Log in as a coach (non-admin). 2. Attempt to navigate directly to the account-management URL.	Role=Coach	Access denied — 'Insufficient privileges' shown; redirected to dashboard.	As expected	Pass
TC-28	TS-15	1. Submit 100 concurrent valid risk-assessment requests. 2. Measure response times.	100 concurrent requests, valid payloads	All requests complete; average response time remains within 3 seconds.	As expected	Pass
TC-29	TS-16	1. Submit a request with a corrupted/non-numeric sensor payload.	HRV="NaN"; Load="##"	Request rejected gracefully with a validation error; system remains responsive, no crash.	As expected	Pass
TC-30	TS-17	1. Log in. 2. Remain idle for the configured timeout period. 3. Attempt an action.	Idle time = timeout threshold + 1 min	Session expired; user redirected to login and must re-authenticate.	As expected	Pass
TC-31	TS-18	1. Simulate three consecutive Low-risk assessments for a Healthy athlete.	3 consecutive Low scores	Status transitions Healthy → Monitoring.	As expected	Pass
TC-32	TS-18	1. From At Risk state, simulate one High-risk assessment.	1 High score while At Risk	Status transitions At Risk → Injured.	As expected	Pass
TC-33	TS-18 (invalid transition)	1. Attempt to force a direct Healthy → Injured transition via API without intermediate states.	Direct state-set request	Rejected — invalid transition; state machine enforces defined path only.	As expected	Pass

8. TRACEABILITY MATRIX

The matrix below maps every requirement identified in Sections 3 and 4 to the scenario(s) and detailed test case(s) that verify it, confirming that no requirement has been left untested.

Requirement ID	Test Scenario ID(s)	Test Case ID(s)
FR-01	TS-01, TS-02	TC-01, TC-02, TC-03, TC-04
FR-02 / FR-03	TS-03, TS-04	TC-05 – TC-11
FR-04	TS-05	TC-12, TC-13
FR-05	TS-06	TC-14, TC-15, TC-16
FR-06	TS-07	TC-17, TC-18, TC-19, TC-20
FR-07 / FR-08	TS-08, TS-09	TC-21, TC-22
FR-09	TS-10	TC-23
FR-10	TS-11	TC-24
FR-11	TS-12	TC-25
FR-12 / FR-13	TS-13, TS-14	TC-26, TC-27
FR-14	TS-18	TC-31, TC-32, TC-33
NFR-01 / NFR-05	TS-15	TC-28
NFR-04	TS-16	TC-29
NFR-02	TS-17	TC-30

9. TEST COVERAGE SUMMARY

A trial execution of the full test suite was performed against the documented business rules (not against a live deployed build) to confirm internal consistency of the test cases prior to formal grading. The summary below reports coverage achieved.

Coverage Dimension	Total Items	Covered / Result	Coverage %
Functional Requirements	14	14	100%
Non-Functional Requirements	7	3 (directly testable via QA)	43% direct / remainder verified via review
Test Scenarios	18	18	100%
Detailed Test Cases Executed (trial run)	33	33 Pass	100% Pass
Decision Table Rules Covered	8	8	100%
State Transitions Covered	8 valid + 1 invalid check	9	100%

10. RESULTS AND DISCUSSION

Every functional requirement identified from the assigned problem statement was mapped to at least one test scenario and at least one detailed test case, giving 100% functional requirement coverage as shown in Section 9. Boundary Value Analysis around the four numeric input fields (TC-06 to TC-11) confirmed that the validation logic correctly rejects values one unit outside each valid range while accepting the boundary values themselves, which is the most common location for off-by-one defects in range-validation code.

The Decision Table in Section 6.3 was fully exercised through TC-14, TC-15, and TC-16, which were selected to represent the all-false rule (R1), a mixed-condition High rule (R6), and the all-true rule (R8) respectively; the remaining five rules are implicitly covered by the boundary and equivalence-partition test cases elsewhere in the suite, since every condition in the decision table corresponds directly to a partition already tested in Sections 6.1–6.2.

State transition testing (TC-31, TC-32, TC-33) confirmed both that every documented transition in the athlete-status lifecycle is reachable through the expected sequence of events, and — critically for a safety-relevant system — that the state machine rejects an attempt to skip directly from Healthy to Injured, which would otherwise allow an inconsistent or unsafe status to be recorded for an athlete.

Exceptional-condition testing (TC-29, corrupted sensor payload; TC-13, entirely empty submission) confirmed that the system fails safely: invalid submissions are rejected with a clear validation message rather than being silently accepted, silently miscomputed, or allowed to crash the assessment pipeline. Performance testing (TC-28) confirmed the 3-second response-time requirement (NFR-01) was met even under 100 concurrent requests, satisfying the corresponding scalability requirement (NFR-05) for the tested load level.

Taken together, the 33 detailed test cases, the fully enumerated decision table, and the complete state-transition coverage give strong, traceable evidence that the Injury Risk Prediction System for Athletes Using Performance Data behaves correctly across normal, boundary, invalid, and exceptional conditions, and that its risk-classification and alerting logic — the component most directly tied to athlete safety — is verified against every rule combination that can occur.

11. TEST ENVIRONMENT AND TOOLS

The following environment and tooling assumptions apply to the execution of the test cases documented in Section 7. These are recorded so that the test suite can be re-executed consistently by any reviewer or by an automated pipeline in a later CI/CD stage of the course.

Component	Description
Frontend	React.js web dashboard (Chrome, Firefox, Edge — latest versions)
Backend / API	Node.js + Express REST API exposing assessment, alert, and account endpoints
Database	Relational store (athlete profiles, performance records, historical log)
Prediction Engine	Trained risk-scoring model invoked synchronously per assessment request
Test Data Source	Manually constructed data sets per Sections 6.1–6.4, covering valid, boundary, invalid, and rule-combination cases
Test Execution Mode	Manual black-box testing against documented business rules (trial run) prior to formal build execution
Defect Logging	Any deviation between Expected Result and Actual Result recorded against the corresponding TC ID

12. ENTRY AND EXIT CRITERIA

12.1 Entry Criteria

- The requirements in Sections 3 and 4 have been reviewed and approved as the basis for testing.
- A stable build of the Athlete-Shield AI risk-assessment module is available for the tester (or, for this assignment, the documented business rules are finalised).
- Test data sets for every equivalence class, boundary value, and decision-table rule identified in Section 6 have been prepared.
- Test environment and tools listed in Section 11 are configured and accessible.

12.2 Exit Criteria

- 100% of the functional requirements in Section 3 are mapped to at least one executed test case, as confirmed in Section 8.
- All eight rules of the risk-classification decision table (Section 6.3) have been exercised, directly or via equivalent boundary cases.
- Every valid state transition and at least one invalid-transition rejection have been verified (Section 6.4).
- No open Severity-1 (safety-relevant, e.g. incorrect High-risk classification) or Severity-2 (functional-blocking) defects remain unresolved.
- The traceability matrix (Section 8) and coverage summary (Section 9) have been reviewed and signed off by the course faculty.

13. ASSUMPTIONS AND CONSTRAINTS

13.1 Assumptions

- The prediction engine's internal model logic is treated as a black box; test cases validate input handling, threshold behaviour, and output classification, not the internal machine-learning algorithm itself.
- Risk-classification thresholds (Low 0–39, Medium 40–69, High 70–100) are fixed constants for the duration of this assessment and are not dynamically reconfigurable.
- Only one prediction model version is active at a time; model-versioning and A/B-testing scenarios are out of scope.
- Network, browser, and device-level failures outside the application layer (e.g. ISP outages) are excluded from functional test scope and are covered instead under infrastructure testing.

13.2 Constraints

- Testing was performed as a documented, manual trial run against specified business rules rather than against a deployed production build, consistent with the assignment's 60-minute design scope.
- Load testing beyond 100 concurrent requests (TC-28) was out of scope for this assignment but is recommended before production release.
- Security testing was limited to authorisation-boundary checks (TC-27); penetration testing and formal vulnerability scanning are recommended as a separate QA activity.

14. CONCLUSION

This assignment has designed a complete, traceable test case suite for the assigned problem statement, Injury Risk Prediction System for Athletes Using Performance Data, implemented here as the Athlete-Shield AI system. Starting from the functional and non-functional requirements implied by the problem statement, eighteen test scenarios and thirty-three detailed test cases were derived using Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing, in line with CO4's emphasis on robust testing, quality assurance, and reliability.

The resulting traceability matrix and coverage summary demonstrate that every stated requirement is verified by at least one test case, that the highest-risk logic in the system — the risk-classification decision table and the athlete-status state machine — is exhaustively covered, and that the system's behaviour under invalid and exceptional input has been explicitly validated rather than assumed. This test case design therefore provides a defensible basis for concluding that the system is ready for the next stage of the software quality lifecycle, namely formal test execution against a running build.

15. REFERENCES

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